

$$① \sin x \sim x$$

$$\tan x \sim x \quad \arctan x \sim x \quad \arcsin x \sim x$$

等价无穷小

$$② \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$$

$$\text{设 } \frac{1}{x} = t, x \rightarrow \infty, t \rightarrow 0, x = \frac{1}{t}$$

$$\lim_{t \rightarrow 0} (1+t)^{\frac{1}{t}} = e \Rightarrow \lim_{t \rightarrow 0} \frac{\ln(1+t)}{t} = 1$$

同边取对数

有 $\ln(1+x) \sim x$

$$③ e^x - 1 \sim x$$

$$\text{设 } t = e^x - 1 \quad \lim_{x \rightarrow 0} e^x - 1 \Rightarrow \lim_{t \rightarrow 0} \frac{\ln(1+t)}{t} = 1$$

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$$

$$④ (1+x)^a - 1 \sim ax$$

$$\Rightarrow e^{\ln(1+x)^a} - 1 \sim a \ln(1+x) \sim ax$$

$$⑤ a^x - 1 \sim x \ln a$$

$$\Rightarrow a^x - 1 \sim x \ln a$$

$$a^x - 1 \Rightarrow e^{\ln a^x} - 1 \sim x \ln a$$

$$⑥ 1 - \cos x \sim \frac{x^2}{2}$$

$$1 - \cos x = 2 \sin^2 \frac{x}{2}$$

常用麦克劳林展开式

$$e^x \sim 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}$$

$$\ln(1+x) \sim x - \frac{x^2}{2!} + \frac{x^3}{3!}$$

$$\sin x \sim x - \frac{x^3}{3!} + \frac{x^5}{5!}$$

$$\sqrt{1+x} - 1 \sim \frac{x^2}{2}$$

$$\cos x \sim 1 - \frac{x^2}{2!} + \frac{x^4}{4!}$$

$$\frac{1}{1-x} \sim 1 + x + x^2$$

$$\tan x \sim x + \frac{x^3}{3}$$

$$\arctan x \sim x - \frac{x^3}{3} + \frac{x^5}{5}$$

莱布尼茨公式

$$(uv)^{(n)} = \sum_{k=0}^n \binom{n}{k} u^{(k)} v^{(n-k)}$$

$$(uv)^{(2)} = u''v + 2u'v' + uv'' = \binom{2}{0} uv'' + \binom{2}{1} u'v' + \binom{2}{2} u''v$$

$$\binom{n}{k} = C_n^k = \frac{n!}{k!(n-k)!}$$

$$\binom{n}{0} = C_n^0 = \frac{n!}{0!n!} = 1$$

例 $f(x) = x^2 \cdot 2^x$, 求 $x=0$ 时 $f^{(n)}(0)$

$$\binom{2}{1} = \frac{2!}{1!1!} = 2$$

$$f^{(n)}(x) = \binom{n}{0} (x^2) (2^x)^{(n)} + \binom{n}{1} (2x) (2^x)^{(n-1)} + \binom{n}{2} (2) (2^x)^{(n-2)} + \binom{n}{3} (x^2)^{(3)} (2^x)^{(n-3)} + \dots$$

$$f^{(n)}(0) = 0 + 0 + \binom{n}{2} \cdot 2 \cdot (2^x)^{(n-2)} = C_n^2 \cdot 2 \cdot (2^x)^{(n-2)} \Big|_{x=0}$$

$$= \frac{n(n-1)}{2} \cdot 2 \cdot 2^x \ln 2^{n-2} = n(n-1) \cdot 2^x \ln 2^{n-2}$$

3 阶及之后为 0

$$(2^x)' = 2^x \ln 2$$

$$(2^x)^{(n)} = 2^x \ln 2^n \quad \text{即} \quad (2^x)^{(n-2)} = 2^x \ln 2^{n-2}$$

$$C_n^2 = \frac{n!}{2!(n-2)!} = \frac{n \cdot (n-1)}{2}$$

例 $f(x) = x^2 \ln(1+x)$, 当 $n \geq 3$ 时, $f^{(n)}(0) =$

$n \geq 3$

$$f^{(n)}(x) = \binom{n}{0} x^2 \ln(1+x)^{(n)} + \binom{n}{1} (2x) \cdot \ln(1+x)^{(n-1)} + \binom{n}{2} 2 \ln(1+x)^{(n-2)} + \binom{n}{3} 0 + \dots$$

$$f^{(n)}(0) = \binom{n}{2} 2 \cdot \ln(1+x)^{(n-2)} \Big|_{x=0} = -C_n^2 2 (n-3)! = -n(n-1) \cdot (n-3)! \quad \ln(1+x)^{(n)} = \frac{-(n-1)!}{(1+x)^n}$$

$$= \frac{-n!}{n-2}$$

相切: $f(x) \nearrow g(x) \rightarrow f'(x) = g'(x)$

曲率相等: $K = \frac{|y''|}{(1+(y')^2)^{3/2}}$, 有 $f''(x) = g''(x)$

且 $\lim_{x \rightarrow a} \frac{f(x) - g(x)}{(x-a)^2} = 0$, 设 $x-a=h, x \rightarrow a, h \rightarrow 0$

$$\lim_{h \rightarrow 0} \frac{f(h+a) - g(h+a)}{h^2} = 0$$

即 $f(a) = g(a)$, 有交点.

根据洛必达法则, $\lim_{h \rightarrow 0} \frac{f'(h+a) - g'(h+a)}{2h} = 0$

$f'(a) = g'(a)$ 相切

再次洛必达 $\lim_{h \rightarrow 0} \frac{f''(h+a) - g''(h+a)}{2} = 0$

$f''(a) = g''(a)$, 曲率相等

条件 $\lim_{h \rightarrow 0} \frac{f(h+a) - g(h+a)}{h^2} = 0$

↓

结果 相切 且 曲率相等

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曲率只要 2 阶导数 绝对值相等

$$\frac{|f''(x)|}{(1+(f'(x))^2)^{3/2}} = \frac{|g''(x)|}{(1+(g'(x))^2)^{3/2}}$$

所以是 充分不必要

不定积分的定义 $\rightarrow f(x)$ 原函数的全体.

$$\int f'(x) dx = f(x) + C$$

$$(F(x)+C)' = f(x) \quad \int f(x) dx = F(x) + C$$

$$\int 1 d(F(x)) = \int t' dt = t + C = F(x) + C$$

整体看作 t

$$\frac{d}{dx} (F(x)+C) = \frac{d}{dx} (\int f(x) dx) = f(x)$$

(求导)

$$\frac{d}{dx} (\int f'(x) dx) = f(x)$$

第一换元积分法

$$d(F(x)+C) = f(x) dx$$

(求微分)

$$\int f(u) du \rightarrow \int f(\varphi(x)) d\varphi(x) \rightarrow \int f(\varphi(x)) \varphi'(x) dx$$

设 $u = \varphi(x)$

$$\text{例: } \int 2\cos x dx \rightarrow \int \cos 2x d2x = \sin 2x + C$$

不定积分的性质

第二换元积分法

$$\int (f(x) \pm g(x)) dx = \int f(x) dx \pm \int g(x) dx$$

$$\int f(x) dx \rightarrow \int f(\varphi(t)) d\varphi(t) = \int f(\varphi(t)) \varphi'(t) dt$$

$$\int (k f(x)) dx = k \int f(x) dx$$

设 $x = \varphi(t)$

$$\text{例: } \int \sqrt{a^2 - x^2} dx, (a > 0)$$

$$a^2 - x^2 \geq 0 \quad a^2 \geq x^2$$

$$\text{假设 } x = a \sin t, \quad x \in (-a, a), \quad -\frac{\pi}{2} < t < \frac{\pi}{2}$$

$$\int \sqrt{a^2 - a^2 \sin^2 t} \cdot a \cos t dt$$

$$= \int a \sqrt{1 - \sin^2 t} \cdot a \cos t dt$$

$$= \int a^2 \cos^2 t dt = a^2 \int \frac{\cos 2t + 1}{2} dt$$

$$= \frac{a^2}{4} \int \cos 2t dt + \frac{a^2}{2} t = \frac{a^2}{4} \sin 2t + \frac{a^2}{2} t + C$$

$$\int \cos^2 t dt = \frac{\sin 2t}{4} + \frac{t}{2} + C$$

二倍角公式

$$\cos(2+\beta) = \cos 2 \cos \beta - \sin 2 \sin \beta$$

$$\cos 2t = \cos^2 t - \sin^2 t$$

$$\cos^2 t = \cos 2t + \sin^2 t$$

$$= \cos 2t + (1 - \cos^2 t)$$

$$2\cos^2 t - 1 = \cos 2t$$

$$\cos^2 t = \frac{\cos 2t + 1}{2}$$

分部积分法

$$\int u dv = uv - \int v du$$

$$\text{推导: } (uv)' = u'v + uv' \rightarrow uv = \int v du + \int u dv$$

$$u' dx = du$$

$$v' dx = dv$$

$$\int (uv)' dx = \int (u'v + uv') dx$$

$$\frac{d}{dx} \int (uv)' dx = \frac{d}{dx} \int u'v dx + \frac{d}{dx} \int uv' dx$$

$$= \int u'v dx + \int uv' dx$$

$$uv = \int v du + \int u dv$$